

Distributed Dynamic Egress Routing with IoT Devices

This document summarizes a distributed approach for computing safe egress paths in a building represented as a dynamic graph, where hazards such as fire or smoke change edge and node costs over time. The focus is on fully distributed, IoT-based systems with local sensing, local communication, and local actuation.

1. Problem Setting

- The building is modeled as a graph with ~50 nodes (rooms, corridors, junctions).
- Edges represent doors or hallway segments.
- Exits are known and static.
- Hazards dynamically modify traversal costs or disable edges.
- Each node is an IoT device with sensors (smoke, temperature, visibility) and actuators (directional emergency lights).

2. Chosen Algorithmic Paradigm

The most suitable approach is distributed value diffusion, equivalent to asynchronous Bellman–Ford or value iteration. Each node maintains a scalar value $V(x)$, representing the current best cost-to-exit from that node.

Update rule at node x :

$$V(x) \leftarrow p(x) + \min_{y \in N(x)} (c(x,y) + V(y))$$

where $p(x)$ is a local risk penalty and $c(x,y)$ is the dynamic edge cost.

3. Information Requirements

Each node requires only:

- Knowledge of its immediate neighbors
- Locally sensed hazard information
- Periodic or event-driven neighbor value updates

No node needs global topology knowledge. Global shortest paths emerge implicitly through information propagation.

4. Optimality Guarantees

Under the assumptions of non-negative additive costs and a temporarily frozen hazard state, distributed value diffusion converges to the exact global shortest-path solution. At convergence, no centralized algorithm with full topology knowledge can produce a better path.

5. Transient (Non-Optimal) Phase

Before convergence, nodes may act on incomplete information. After k communication rounds, each node has the optimal path among all paths of $\leq k$ hops.

For a well-connected 50-node indoor graph:

- Typical hop-diameter to an exit: 6–8 hops

- With 1 s update period: convergence in ~6–8 s
- Faster with event-driven hazard broadcasts

6. Bounding Suboptimal Behavior

If each edge cost is bounded by $c_{\max}$ and information propagation delay is k rounds, then worst-case extra cost before correction is approximately $k \times c_{\max}$. In practice, communication is much faster than human movement, so incorrect guidance typically persists for only one decision point.

7. Actuation and Stability

Emergency lights point toward the neighbor minimizing $c(x,y) + V(y)$. To ensure stability and human trust, systems should include:

- Hysteresis (minimum improvement threshold)
- Minimum hold times for directions
- Sensor smoothing and confidence aging

8. Fault Tolerance

The distributed approach is resilient by design. Failed nodes or links are treated as unreachable. Stale neighbor information can be penalized, naturally biasing routes toward well-instrumented and recently updated regions.

9. Key Takeaways

- Global topology knowledge is not required for optimal routing.
- Optimality is achieved through convergence, not centralized planning.
- The main engineering challenge is minimizing and managing the transient phase.
- Distributed value diffusion aligns well with IoT constraints and safety needs.